Huggingface: Fine-Tuning

What is Fine-tuning

- Fine-tuning is the process of taking a powerful, pre-trained model and further training it on your specific dataset.
- This makes the generalist model an expert for your particular task.

Fine-tuning Process

- **Prepare the Data:** Load a dataset and a matching tokenizer. Tokenize the entire dataset so it's in a format the model can understand.
- **Load the Model:** Load a pre-trained model with a task-specific head (e.g., AutoModelForSequenceClassification).
- **Define Training Arguments:** Create a configuration that specifies all the hyperparameters for training, like learning rate, number of epochs, batch size, and where to save the model.
- **Define Metrics:** Create a function that the Trainer can use to calculate evaluation metrics like accuracy during training.
- **Train:** Instantiate the Trainer with all the components above and call the .train() method.

Huggingface Trainer API

- It takes your data, creates batches of the size you specified (per_device_train_batch_size).
- For each batch, it performs a forward pass: inputs go through the model to produce logits.
- It calculates the loss.
- It performs backpropagation.
- It repeats this for all batches to complete an epoch.
- At the end of each epoch it runs your compute_metrics function on the evaluation set to report the accuracy.
- It saves checkpoints of your model periodically to the output_dir.
- Once training is finished, your fine-tuned model, tokenizer, and configuration files will be saved in the directory you specified.

Huggingface Trainer API

- The Trainer API provides a high-level interface that handles most training complexity
- Use processing_class to specify your tokenizer for proper data handling
- TrainingArguments controls all aspects of training: learning rate, batch size, evaluation strategy, and optimizations
- compute_metrics enables custom evaluation metrics beyond just training loss
- Modern features like mixed precision (fp16=True) and gradient accumulation can significantly improve training efficiency